

A unit-coefficient counterexample to 00000001198

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10 October 2026

The conjecture says that Zonal Littlewood–Richardson coefficients are nonnegative alternating sums of even integers. Its English and Chinese statements do not exclude the empty partition. That coefficient alone contradicts the claimed parity restriction.

The actual polynomial and normalization. Write Z_λ for the normalized Zonal polynomials, that is, the Jack parameter-two specialization. The empty partition $\emptyset$ has weight zero. Its normalized polynomial is

$$Z_\emptyset(x_1, \dots, x_m) = 1.$$

This is the actual degree-zero symmetric polynomial, rather than an auxiliary scalar model. For the standard Zonal normalization, the NIST Digital Library of Mathematical Functions, equations 35.4.4 and 35.4.6, explicitly include the zero partition. The latter identity at degree zero has exactly one summand and reads $Z_\emptyset(T) = (\text{tr } T)^0 = 1$. Equivalently, in monic Jack normalization the zero partition has the single monomial $m_\emptyset = 1$ and no lower terms. The usual integral normalization multiplies this by an empty product, also 1. Thus the example is independent of these standard normalization choices.

Define the LR coefficients by actual multiplication and expansion in this homogeneous basis:

$$Z_\lambda Z_\mu = \sum_{|\nu|=|\lambda|+|\mu|} c_{\lambda\mu}^\nu Z_\nu.$$

When $\lambda = \mu = \emptyset$, the only weight-zero partition is $\nu = \emptyset$. Consequently

$$1 = Z_\emptyset^2 = c_{\emptyset\emptyset}^\emptyset Z_\emptyset, \quad c_{\emptyset\emptyset}^\emptyset = 1.$$

The coefficient is unique: extraction of the constant monomial coefficient from the polynomial equality gives $1 = c$.

Theorem 1. *The parity assertion in 00000001198 is false.*

Proof. For any integers $a_0, \dots, a_{r-1}$,

$$\sum_{j=0}^{r-1} (-1)^j (2a_j) = 2 \sum_{j=0}^{r-1} (-1)^j a_j$$

is even. The LR coefficient above is 1, which is odd, so it is no such alternating sum. Requiring the sum to be nonnegative does not remove this obstruction. Allowing arbitrary signs instead of strictly alternating signs also preserves evenness. $\square$

Scope and Lean verification. This disproves the stated even-integer clause, without asserting that Zonal LR positivity or any corrected matching interpretation is false. The source imposes no positive-weight restriction. The Lean project uses actual multivariate polynomials over $\mathbb{Q}$ in an arbitrary variable type. It constructs $Z_\emptyset$ as the zero-exponent monomial with coefficient 1, proves permutation invariance, proves that every degree-zero polynomial lies in the

constant basis, and extracts the unique LR unit coefficient from multiplication. It then proves the alternating-sum parity obstruction by induction on a list of integers. All printed axiom audits use only the permitted standard Lean axioms. The project pins Lean 4.19.0 and mathlib commit `c44e0c8ee63ca166450922a373c7409c5d26b00b`.